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Unit “ 3 : Session “ 11 : SLO “ 1 | Circular Queue “ Introduction

■ I. Fill in the Blanks

- A circular queue connects the LAST position back to the front.
- Circular queue overcomes the problem of MEMORY WASTAGE (unused spaces) in a linear queue.
- In a circular queue, when the rear reaches the end, it wraps around to index 0.
- A circular queue follows the FIFO principle.
- The condition for a full circular queue is: $(\text{rear} + 1) \% \text{size} == \text{FRONT}$.
- The condition for an empty circular queue is: $\text{front} == -1$.
- When the first element is inserted, both front and rear point to the SAME (0th) index.
- Circular queues are commonly used in ROUND-ROBIN scheduling.
- The ISFULL function checks if a circular queue is full.
- The ISEMPY function checks if a circular queue is empty.
- Circular queues help in utilizing memory more EFFICIENTLY than linear queues.
- In a circular queue, incrementing the rear is done using MODULO (%) operator.
- Circular queues can be implemented using arrays or LINKED LISTS.
- The front element of a circular queue is accessed using the FRONT pointer.
- In circular queues, rear and front move in a CIRCULAR (wrap-around) manner.